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Roll No.

ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course: B.Tech (C.S.E)

Semester: III

Name of the Paper: Fundamental of Information Security and Blockchain

Paper Code: TCS-332

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	CO1
(a)	Explain the role of white hat, grey hat, and black hat hackers in the context of ethical hacking.	
(b)	Discuss CIA triad. Why it plays a crucial role in Information Security.	
(c)	Why Penetration Testing is important. Discuss the various phases involved in penetration testing.	
Q2	(20 marks)	CO2
(a)	Write Linux Commands for the following tasks. i) Print Current Working Directory ii) Add a user iii) Remove a file iv) Create an Empty File v) Create new Directory	
(b)	Create a Bash script to calculate the factorial of a given number.	
(c)	Write a Python Program to check whether a given string is a palindrome.	
Q3	(20 marks)	CO3
(a)	Briefly Discuss Swiss Army Knife of TCP/IP connections.	
(b)	Explain Cross Site Scripting Attack. What methods can be implemented to prevent XSS attacks in web applications?	
(c)	What is the Difference between Phishing and Spear Phishing? Provided suitable examples.	
Q4	(20 marks)	CO4 + CO5
(a)	Define a blockchain and explain its core characteristics.	
(b)	How Merkle root is used in Payment Verification. Justify your answer with a suitable example.	
(c)	What is the Genesis Block, and why is it important in blockchain networks?	
Q5	(20 marks)	CO5 + CO6
(a)	Compare Proof of Work (PoW) and Proof of Stake (PoS).	
(b)	Explain Below Mentioned Terms:	
	i) Distributed Ledger ii) Smart Contract	
(c)	What is a Merkle Tree? Also mention the key benefits of using Merkle Tree.	

